

Operating Systems

Jo Hallett

April 16, 2026



When I was very small



Then my Dad bought home this



It has been brought to my attention that I have a reputation...



BREAKING: University of Bristol Lecturer Caught Red-Handed Using a Windows Machine!



By Joe Barlow

February 18 2023

In a shocking turn of events, renowned computer science lecturer Joseph Halett of the University of Bristol was caught using a Windows machine instead of his beloved Linux software.

I have some opinions about this...

Firstly:

I'd just like to interject for a moment. What you're referring to as Linux, is in fact, GNU/Linux, or as I've recently taken to calling it, GNU plus Linux. Linux is not an operating system unto itself, but rather another free component of a fully functioning GNU system made useful by the GNU corelibs, shell utilities and vital system components comprising a full OS as defined by POSIX. :-P

I have some more opinions about this...

Secondly:

When asked about the incident, Halett simply stated: "I needed to check my email, and my Linux machine was down for maintenance. I had no choice but to use a Windows machine. I swear, it's a one-time thing."

Is entirely fair.

- ▶ I will ~~give exam answers~~ personally thank any student who gives me a working mbsync config for the University exchange service.

But mostly...

- ▶ My official University computers are *Macs*
- ▶ My personal laptop I write most of my lectures on usually runs:

```
$ uname -a
```

```
OpenBSD bananaslug 7.8 GENERIC#1 amd64
```

Well except....

```
$ uname -a
```

```
Linux matilda 6.19.9-200.fc43.x86_64 #1 SMP PREEMPT_DYNAMIC Thu Mar 19 21:34:56 UTC 2025
```

(Y'see I'm teaching Software Tools... and need to be able to test the VMs
...otherwise I have to use ~~one of those awful~~ those really rather lovely lab machines).
(This is of course actually a lie... I've rediscovered video games and since Steam they mostly
just work)

(My server runs OpenBSD still though...)

Confessions of an OS fan

I <3 operating systems

But when I look round the labs I see most people using Windows

- ▶ A few Macs
- ▶ A few people running Linux in Windows
- ▶ And the odd Linux machine (typically Ubuntu or Arch)

Where have all the weird OSs gone?

So what's the plan?

In this lecture:

- ▶ A bit of a history lesson
- ▶ Go through how different OSs grew
- ▶ Show you some of the weirder OSs
- ▶ and where they came from

None of this is examinable...

In the beginning

There were mainframe computers...

- ▶ Large complicated bits of big iron
 - ▶ (as opposed to measely server tin)
- ▶ You could run a whole business off of one (and we still do)

The earliest computers (1950s) ran programs directly loaded into memory off of punched cards.

- ▶ But after a while we started to think it'd be nice to have some utilities stored
- ▶ Help you read data and load programs automatically
- ▶ So by the 1960s the first OSs started to appear...

Some of the earliest mainframe OSs included IBM's System/360



From the Computer History Museum

Oh and by the way these things still exist...

IBM still sell mainframes.

And the modern z/OS is still compatible with System/360 programs

- ▶ (and Linux)

Why would you use one?

- ▶ Big overpowered server machines
- ▶ With really large and fast numbers of CPUs and memory
- ▶ Good for running whole organisations off of
- ▶ Good for data-center scale analytics



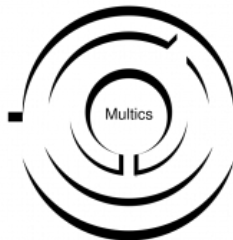
And then...

Dennis Ritchie and Ken Thompson (and others) were working on a mainframe OS called Multics

- ▶ The Multiplexed Information and Computing Service
- ▶ ...first OS to have a shell that'd be recognisable today
- ▶ ...it had a new release in 2023!

Multics was big and complex

- ▶ Project was massively overrunning
- ▶ Bell Labs folks were frankly sick of it
- ▶ ~1970 they decided to write a simpler version for cheaper and more computers
- ▶ Called it Unics: Uniplexed Information and Computing Service



UNIX System V

Unics got renamed Unix by Brian Kernighan

UNIX System 5 (1983) is when things start to get interesting

- ▶ At this stage its a popular OS for many different computers
- ▶ Especially in universities...

IEEE chose it as the basis for their Operating System Standard IEEE 1003 (IEEE-IX)

- ▶ Richard Stallman suggests the name POSIX for it

Operating systems are freely shared

- ▶ Ken Thompson has a sabbatical at Berkeley and helps them install UNIX System 6
- ▶ Together with Bill Joy (vi) starts a distribution of UNIX: BSD

Also in the 1980s

Cheap(er) home micro-computers are starting to become available

- ▶ Typically running off of cassette tapes...
- ▶ But the fancier ones have disk drives and a little persistent storage

Typically run operating systems called Disk Operating Systems or DOS

- ▶ Apple DOS for the Apple][
- ▶ PCDOS for for IBM Personal Computer (later renamed MS-DOS)
- ▶ CP/M for DEC machines
- ▶ C64 DOS for the Commodore 64
- ▶ Atari DOS for the Atari

If you use Powershell or `cmd.exe` in the labs... you're using a descendant of DOS!



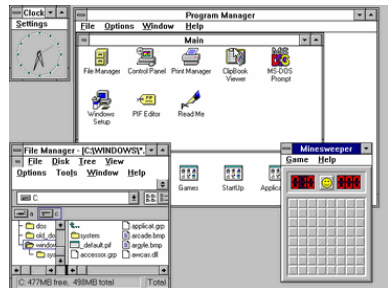
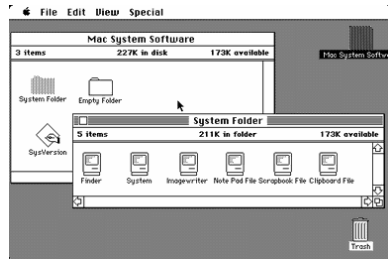
In Cupertino...

Apple make their first Mac OS (System 1) in 1984

- ▶ Licenses Xerox PARCs new graphical user interface and desktop metafores
- ▶ First widely available OS that looks like a modern OS
- ▶ 1985 Steve Jobs gets kicked out of Apple

Microsoft try and copy it for Windows 1.0 (1985)

- ▶ But it's rather clunky
- ▶ But by Windows 3.1 (1991) it's rather good...



Back in UNIX land...

By the early 1990s

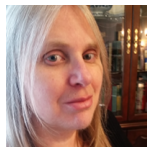
- ▶ Andrew Tannenbaum has a teaching UNIX clone called Minix
- ▶ BSD is up to version 4.3

Copyright is starting to be a thing with computers

- ▶ UNIX owners getting snippy at BSD
- ▶ William and Lynne Jolitz at Berkely do a clean room reimplementation of the more protected bits
- ▶ Call it 386BSD

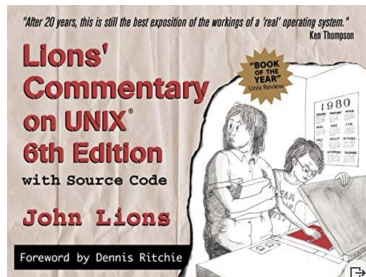
Steve Jobs goes and founds a computer company called NeXT

- ▶ Bases their operating system on 4.3 BSD
- ▶ Calls it NeXTSTEP
- ▶ Initially a commercial failure, but by 1991 starting to be popular in academia
- ▶ Tim Berners-Lee makes the internet for it



Then theres this guy...

UNIX is becoming ubiquitous



Source code is routinely shared and used as part of university OS classes.

Andrew Tannenbaum creates a teaching Microkernel called Minix

- ▶ Students implement bits of the OS for their coursework



Takes the class, creates his own kernel based on Minix

Anyone want to guess what he called it?

He thought about calling it Linux but decided it was to egotistical

- ▶ A colleague who uploaded it to the university's FTP server disagreed and renamed it
- ▶ Linus accepted that its now called Linux

It starts to get popular as a free UNIX clone

GNU userland ported to it until they can build their own kernel (HURD)

- ▶ They never do

Andrew Tannenbaum says it's already obsolete compared to Minix

Back in Cupertino

Apple without Steve doesn't do too well

- ▶ Apple spends its last pennies buying NeXT, under the agreement that Steve will be the new CEO
- ▶ Steve cancels the clones and weird projects

Starts the process of replacing Mac OS System 7/8 with a new OS that merges the work NeXT had been doing

- ▶ This will become MacOS X
- ▶ Ever done any Mac programming and wondered why all the classes descend from an NSObject?



First iMac comes out

- ▶ My Dad brings home a Bondi Blue iMac
- ▶ Had a built in modem and came with a Demon Internet sign up
- ▶ No floppy disks... but new Iomega Zip drives

Still the most beautiful computer ever

- ▶ I completely fell in love
- ▶ Terrible mouse though



MacOS X/macOS

Apple switches from the PowerPC architecture to i386 around 2000

- ▶ Mac OS System 9 becomes Mac OS 10... branded Mac OS X

Looks like a Mac

- ▶ Internally its NeXTStep with the Mach kernel from MIT (another UNIX clone)
- ▶ Userland from FreeBSD (check the man pages!)

NeXTSTEP apps become Mac things

- ▶ Only Chess in your Utilities folder is completely unchanged
- ▶ Some hangovers from Classic Mac OS still there (and a translation layer called Rosetta)

Server version becomes fully POSIX/UNIX compliant in order to deal with an advertising lawsuit

iOS released in 2006...

- ▶ Even more NeXTSTEP-y internally
- ▶ But completely new UI

Microsoft admit their OS is rubbish

Linux continues to grow, especially for servers

- ▶ Many distributions of Linux become available

In 1993 Microsoft adds the POSIX subsystem

- ▶ Then replaces it with Windows Services for UNIX (really BSD)
- ▶ Then replaces it with Windows Subsystem for Linux in 2016

Computer science students everywhere rejoice because they don't have to choose between having an easy life at uni, and having to install a weird OS that won't play games

- ▶ But it still is rubbish and janky

And so we arrive at the current day...

We have **MacOS** that is based on 4.3 BSD

We have **Linux** that is based on a clone of 4.3
BSD

We have **Windows** DOS + a janky Linux-layer



This makes me sad...

Innovation in OS design didn't end in 1986

Most users don't have a choice in what OS they use

- ▶ But we're experts...
- ▶ We should try and use weird systems
- ▶ We should help build new paradigms

It doesn't have to be a monoculture!

The BSDs are still going!

BSD is still out there

386BSD became FreeBSD when merged with BSD 4.4

NetBSD became a fork of 386BSD focussed on supporting more systems

OpenBSD forked NetBSD, audited the code and aggressively removed legacy cruft

DragonflyBSD forked FreeBSD to add a new filesystem

They're just like Linux for the most part

- ▶ But more holistically designed
- ▶ Better documented
- ▶ FreeBSD even has Linux emulation
 - ▶ OpenBSD had it but decided it was cruft, alongside bluetooth

You're probably even using BSD...

Played on a Nintendo Switch? Runs FreeBSD

Played on a PS3/PS4/PS5? Runs FreeBSD

Watched Netflix? All runs off of FreeBSD servers

Using ZFS? It wasn't developed on Linux...

ssh'd into the labs? OpenBSD maintain OpenSSH

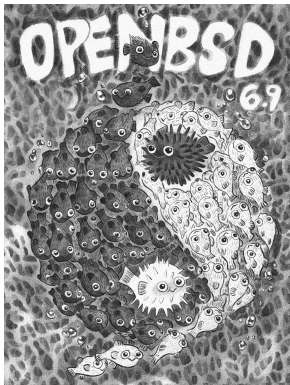
Used tmux? OpenBSD maintain it

Used a firewall on a Mac/PF? OpenBSD code

Using ROP? $W \oplus X$ memory? Stack canaries...? OpenBSD did the first implementations

Try running one next time you fancy trying a new OS!

I use OpenBSD!



I've been using it continuously since OpenBSD 6.9

- ▶ About 5 years now

What I like

- ▶ Really simple
- ▶ Either just works or is never going to work
- ▶ Or I can fix **without swearing** by looking at the code and fixing it
- ▶ No SystemD... just old school shell scripts
- ▶ Man pages contain everything
- ▶ Michael W. Lucas books
- ▶ Doesn't get in my way
- ▶ Fixes stupid design decisions

What I don't like

- ▶ Slow
- ▶ Virtualization sucks
- ▶ Some software missing (Dyalog APL, Signal, Torch, any games...)
- ▶ Fixing stupid design decisions breaks things occasionally

Suppose I want to restrict an app to just using the safe stuff from libc

Linux has Seccomp BPF for that!

```
static int install_filter(void) {
    struct sock_filter filter[] = {
        /* Grab the system call number */
        BPF_STMT(BPF_LD+BPF_W+BPF_ABS, syscall_nr),
        /* Jump table for allowed syscalls */
        BPF_JUMP(BPF_JMP+BPF_JEQ+BPF_J, __NR_rt_sigreturn, 0, 1),
        BPF_STMT(BPF_RET+BPF_K, SECCOMP_RET_ALLOW),
#ifdef __NR_sigreturn
        BPF_JUMP(BPF_JMP+BPF_JEQ+BPF_K, __NR_sigreturn, 0, 1),
        BPF_STMT(BPF_RET+BPF_K, SECCOMP_RET_ALLOW),
#endif
        BPF_JUMP(BPF_JMP+BPF_JEQ+BPF_K, __NR_exit_group, 0, 1),
        BPF_STMT(BPF_RET+BPF_K, SECCOMP_RET_ALLOW),
        BPF_JUMP(BPF_JMP+BPF_JEQ+BPF_K, __NR_exit, 0, 1),
        BPF_STMT(BPF_RET+BPF_K, SECCOMP_RET_ALLOW),
        BPF_JUMP(BPF_JMP+BPF_JEQ+BPF_K, __NR_read, 0, 1),
        BPF_JUMP(BPF_JMP+BPF_JEQ+BPF_K, __NR_write, 0, 1),

        /* Check that read is only using stdin */
        BPF_STMT(BPF_LD+BPF_W+BPF_ABS, syscall_arg(0)),
        BPF_JUMP(BPF_JMP+BPF_JEQ+BPF_K, STDIN_FILENO, 4, 0),
        BPF_STMT(BPF_RET+BPF_K, SECCOMP_RET_KILL),
    }
```


We're not done yet!

```
/* Check that write is only using stdout */
BPF_STMT(BPF_LD+BPF_W+BPF_ABS, syscall_arg(0)),
BPF_JUMP(BPF_JMP+BPF_JEQ+BPF_K, STDOUT_FILENO, 1, 0),
/* Trap attempts to write to stderr */
BPF_JUMP(BPF_JMP+BPF_JEQ+BPF_K, STDERR_FILENO, 1, 2),

BPF_STMT(BPF_RET+BPF_K, SECCOMP_RET_ALLOW),
BPF_STMT(BPF_RET+BPF_K, SECCOMP_RET_TRAP),
BPF_STMT(BPF_RET+BPF_K, SECCOMP_RET_KILL),
};

struct sock_fprog prog = {
    .len = (unsigned short)(sizeof(filter)/sizeof(filter[0])),
    .filter = filter,
};

prctl(PR_SET_NO_NEW_PRIVS, 1, 0, 0, 0);
prctl(PR_SET_SECCOMP, SECCOMP_MODE_FILTER, &prog);
return 0;
}
```

Seccomp-BPF is an entire programming language enabling limited code to be run in the kernel

And in OpenBSD land...

```
void install_filter(void) {  
    pledge("stdio");  
}
```

Pledge restrict program features

Unveil restrict program paths

To be fair Linux's Seccomp BPF is a lot more powerful than OpenBSD's pledge and unveil...

- ▶ But sometimes you just want to secure the app
- ▶ And sometimes you don't want to have to write C pseudo assembly for a firewall to do it

The Bell Labs team didn't retire after UNIX...

What do you do after you invent the OS that defines the modern world?

- ▶ You invent a new better OS!

Plan 9

- ▶ Released to research in 1992, and the public in 1995
- ▶ EVERYTHING is a file
- ▶ Rob Pike and Ken Thompson invent UTF-8 for it (on the back of a napkin)
- ▶ Terminals replaced with a GUI
- ▶ Designed for distributed systems (the cloud before we called it that)

Ideas from Plan 9 have appeared in modern OSs

- ▶ `/proc /sys` debugfs filesystems to allow configuration and monitoring of processes and the kernel through files
- ▶ Go compiler infrastructure is Plan 9's (Rob Pike built both!)

Go to www.9front.org for a modern distribution

- ▶ Beware: lots of old school CS "humour" and memes
- ▶ MNT Reform laptop officially supports it?!

For example...

How do you read where the mouse is?

On Linux via an ioctl

On Plan 9 `cat /dev/mouse`

How do you draw to the screen?

On Linux More ioctl magic!

On Plan 9 `Open /dev/draw/new`

How do you find information about a process

On Linux In `/proc`

On Plan 9 ...where'd you think Linux copied this feature from? ;-)

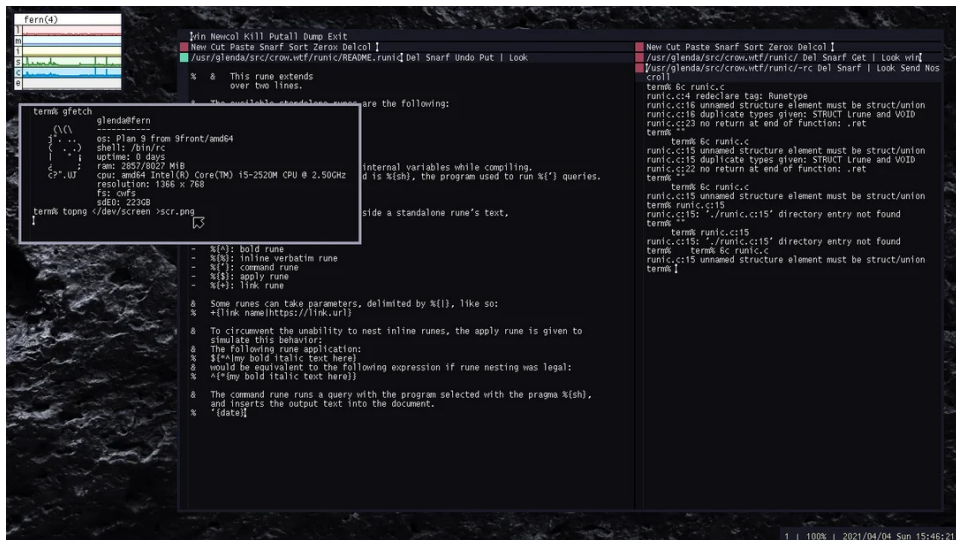


ACME and Plan 9

win Newcol Kill Putall Dump Exit	New Cut Paste Snarf Sort Zerox Delcol
/usr/maht/~gr Del Snarf Look Send Noscroll	Errors Del Snarf Look
term% ftps ftp.proweb.co.uk	NAME
220 Welcome to Proweb Web Server (Jerry)	acme, win, awd - interactive text windows
331 Please specify the password.	SYNOPSIS
230 Login successful.	acme [-ab] [-c ncol] [-f varfont] [-f fixfont] [-l loadfile] [file ...]
215 UNIX Type: L8	win [command]
257 "/"	awd [label]
term% cat /dev/window topng > /n/ftp/www/acme_screeny.png	DESCRIPTION
[Del Snarf Undo Look	Acme manages windows of text that may be edited interactively or by external programs. The interactive interface uses the keyboard and mouse; external programs use a set of files served by acme; these are discussed in acme(4).
This image is in the Public Domain]	Any named files are read into acme windows before acme accepts input. With the -l option, the state of the entire system is loaded from loadfile, which should have been created by a Dump command (q.v.), and subsequent file names are ignored. Main files display as text; directories display as columnated lists of the names of their components, as in ls -p directory/mc except that the names of subdirectories have a slash appended.
/usr/maht/ Del Snarf Get Look	The -f (-F) option sets the main font, usually variable-pitch (alternate, usually fixed-pitch); the default is /lib/Font/blt/lucidasans/euro.8.font
386/ freetype-plan9.tgz momo/	(.../lucm/unicode.9.font). Tab intervals are set to the width of 4 (or the value of \$tabstop) numeral zeros in the appropriate font.
abaco.tgz freetype/ radio	Windows
abaco/ hugs-p9/ radios	Acme windows are in two parts: a one-line tag above a multi-line body. The body typically contains an image of a file, as in sam(1), or the output of a program, as in an rio(1) window. The tag contains a number of blank-separated words, followed by a vertical bar character, followed by anything. The first word is the name of the window, typically the name of the associated file or directory, and the other words are commands available in that window. Any text may be added after the bar; examples are strings to search for or commands to execute in that window. Changes to the text left of the bar will be ignored, unless the result is to change the name of the window.
acme.dump hugs.tgz sys/	
acme_screeny.png lib/ tmp/	
bin/ mad.tgz window	
freetype-2.1.4/ madplay/	
/usr/maht/lib/plumbing Del Snarf Look	/n/momo/ Del Snarf Get Look This is my OpenBSD machine pver u9fs
% to update: cp /usr/maht/lib/plumbing /mnt/plumb/rules	cschr bin/ bsd.mp dev/ home/ shini/ usr/
editor = acme	profile boot bsd.rd emul/ mnt/ stand/ var/
include basic	altroot/ bsd cvs/ etc/ root/ tmp/
type is text	/n/momo/etc/motd Del Snarf Look
data matches '(a-zA-Z0-9_\\-./-)' on line ([0-9])-S'	OpenBSD 3.8 (GENERIC) #138: Sat Sep 10 15:41:37 MDT 2005
arg ifile /n/momo/var/www/S1	Welcome to OpenBSD: The proactively secure Unix-like operating system.
data set \$file	
attr add add=\$2	
plumb to edit	
plumb client \$editor	
type is text	
data matches '(a-zA-Z0-9_\\-./-)'(classlinc)	
arg ifile /n/momo/var/www/php/S1	
data set \$file	
plumb to edit	
plumb client \$editor	
type is text	
data matches ->([a-zA-Z0-9_\\-./-])	
plumb start rc -c 'ssh momo "/bin/functionlist.rc" grep \$1' >{2-1}	
plumb -i -d edit	

Available as part of 9ports in most other OSs.

People even theme it now?!



```
fern(4)
1
m
1
5
C
B

Win Newcol Kill Putall Dump Exit
New Cut Paste Snarf Sort Zerox Delcol !
/usr/glanda/src/crow.wtf/runic/README.runic Del Snarf Undo Put | Look

% & This rune extends
over two lines.

The available standalone runes are the following:

term$ gfetch
glenda@fern
-----
i.. os: Plan 9 from 9front/amd64
. . . shell: /bin/rc
. . . uptime: 0 days
. . . ram: 2857/8027 MIB
c?*.w : cpu: amd64 Intel(R) Core(TM) i5-2520M CPU @ 2.50GHz
      resolution: 1366 x 768
      fs: cifs
      sdED: 223GB
term$ topng </dev/screen >scr.png

- %(A): bold rune
- %[%]: inline verbatim rune
- %{}: command rune
- %$: apply rune
- %+: link rune

& Some runes can take parameters, delimited by %{}, like so:
% +{link name|https://link.url}

& To circumvent the inability to nest inline runes, the apply rune is given to
simulate this behavior:
& The following rune application:
% ${^}my bold italic text here}
& would be equivalent to the following expression if rune nesting was legal:
% ^{^}my bold italic text here}

& The command rune runs a query with the program selected with the pragma %(sh),
& and inserts the output text into the document.
% {date}

New Cut Paste Snarf Sort Zerox Delcol !
/usr/glanda/src/crow.wtf/runic/ Del Snarf Get | Look win
/usr/glanda/src/crow.wtf/runic/-rc Del Snarf | Look Send Nos
crol}
term$ 6c runic.c
runic.c:14 redeclare tag: Runetype
runic.c:16 unnamed structure element must be struct/union
runic.c:16 duplicate types given: STRUCT l rune and VOID
runic.c:23 no return at end of function: .ret
term$
term$ 6c runic.c
runic.c:15 unnamed structure element must be struct/union
runic.c:15 duplicate types given: STRUCT l rune and VOID
runic.c:22 no return at end of function: .ret
term$
term$ 6c runic.c
runic.c:15 unnamed structure element must be struct/union
term$ runic.c:15
runic.c:15: './runic.c:15' directory entry not found
term$
term$ runic.c:15
runic.c:15: './runic.c:15' directory entry not found
term$ term$ 6c runic.c
runic.c:15 unnamed structure element must be struct/union
term$ !
```

1 | 100% | 2021/04/04 Sun 15:46:21

(via <http://reddit.com/r/unixporn>)

Qubes OS

A reasonably secure OS

- ▶ "If you're serious about security QubesOS is the best OS available today. It's what I use, and free." — Edward Snowden

Designed by Joanna Rutkowska

- ▶ Based on Xen and Linux sort of...

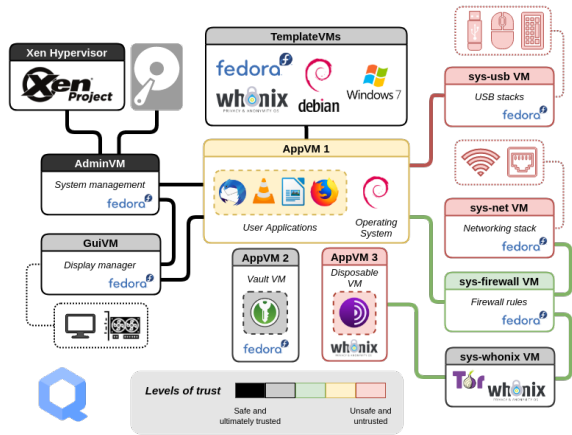
Basic ideas:

- ▶ Virtualise everything
- ▶ Nothing is permanent

<https://www.qubes-os.org>



Qubes Architecture



Downsides...

You need to be able to spawn lots of VMs

- ▶ So somewhat odd hardware requirements
- ▶ Its a bit slower
- ▶ Usability/ricing is an afterthought

But...

- ▶ It feels like something from the future
- ▶ Hot swapping distros in seconds (well... tens of seconds) makes you giddy
- ▶ I wrote my PhD thesis using it : -)

Ideas from Qubes are starting to appear in Linux and iOS

- ▶ Fedora Silverblue implements a **lot** of Qubes's ideas, but in a more usable way

Version 4.3 came out at Christmas!

The Uxn CPU

What if we all die in global thermonuclear war/global warming/out of sheer ennui with modern life?

- ▶ No more making new computers
- ▶ Need to reuse the ones we already have!

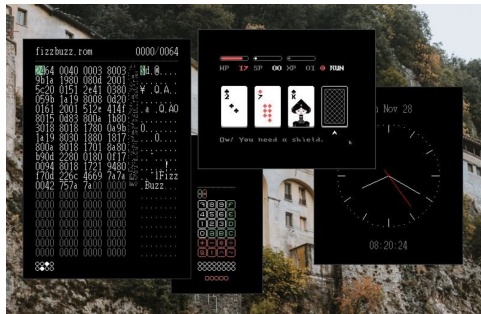
Whats the minimum viable computer we can use until the mutant-tuna enslave us all?

Uxn

Designed by Hundred Rabbits (Rek and Devine) to be an OS that can run on other computers

- ▶ Can run on a 6502 processor (a NES), and fairly easy to port to anything else
- ▶ Stack based vm (Forth programming!)

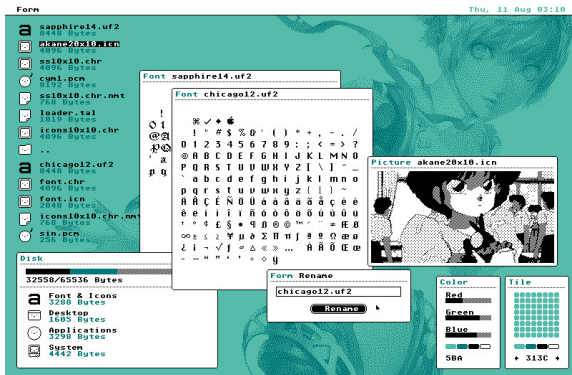
<https://100r.co/site/uxn.html>



Uxn now has an OS

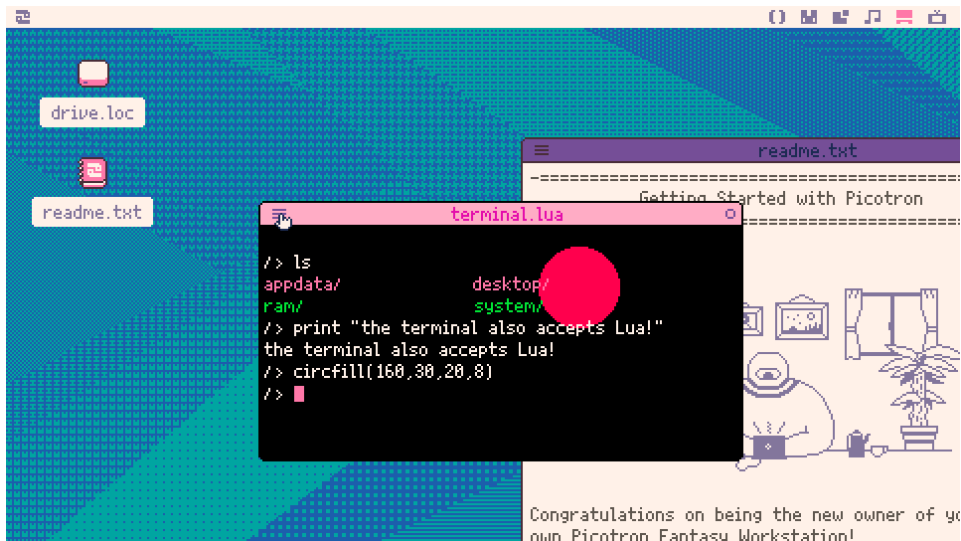
Because the Uxn virtual machine has been ported everywhere... how do you load files on a NES?

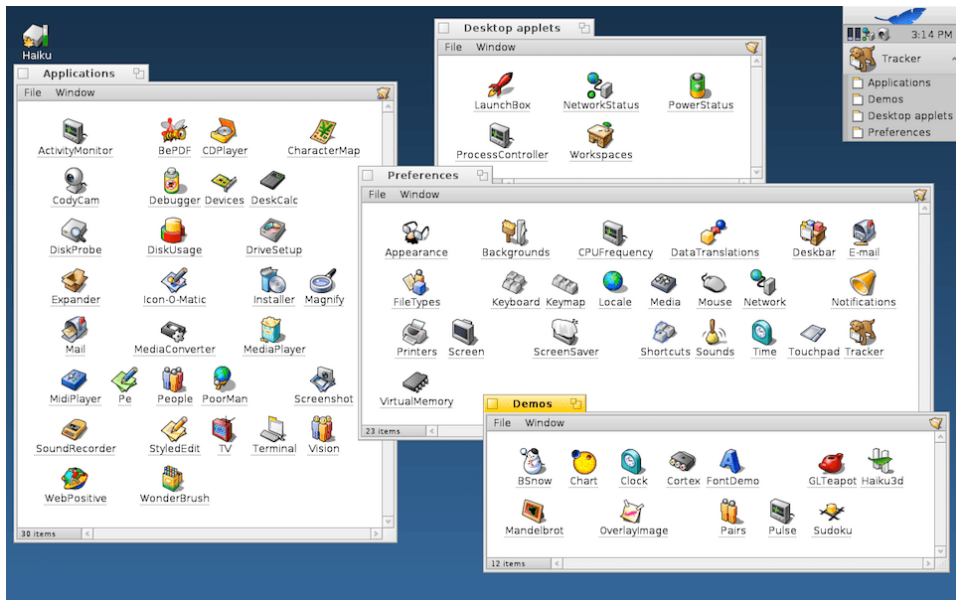
- You could do with a real OS to manage stuff



Potato Uxn is a simple little OS for the Uxn CPU!

Picotron

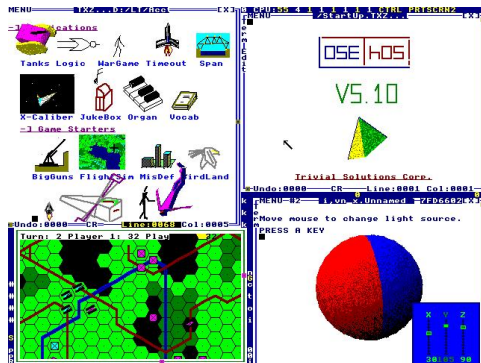




Temple OS

Temple OS is a 64bit OS written by Terry Davis

- ▶ Features its own compiler, filesystem, graphics and UI library
- ▶ Is fully runtime recompilable (like a Lisp machine!)
- ▶ Has 3D games and simulations
- ▶ An immense work of art (and devotion) by 1 person

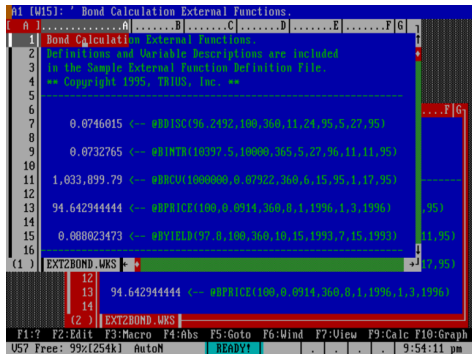


Note: Terry Davis was extremely ill, and died in 2018.

Or try a modern take on an old one...

FreeDOS is a modern DOS...

- ▶ Even has tab-complete support!
- ▶ Word Perfect is now Open Source!



```
A1 [W15]: ' Bond Calculation External Functions.
[ A ] .....B .....C .....D .....E .....F G
1 Bond Calculation External Functions.
2 Definitions and Variable Descriptions are included
3 in the Sample External Function Definition File.
4 ** Copyright 1995, TRIUS, Inc. **
5
6 .....F G
7 0.0746015 (← @BID3C(96.2492,100,360,11,24,95,5,27,95)
8
9 0.07322765 (← @BINTR(10397.5,10000,365,5,27,96,11,11,95)
10
11 1,033,899.79 (← @BDCU(1000000,0.07922,360,6,15,95,1,17,95)
12
13 94.642944444 (← @BPRICE(100,0.0914,360,0,1,1996,1,3,1996)
14
15 0.088023473 (← @BYIELD(97.0,100,360,10,15,1993,7,15,1993)
16
(1) EXT2BOND.WKS +
12
13 94.642944444 (← @BPRICE(100,0.0914,360,0,1,1996,1,3,1996)
14
(2) EXT2BOND.WKS
F1:? F2:Edit F3:Macro F4:Abs F5:Goto F6:Wind F7:View F9:Calc F10:Graph
U57 Free: 99%(254k) AutoM READY! . . . 9:54:11 pm
```

Medley is a modern take on Interlisp for the Lisp Machine

- ▶ Wouldn't it be nice to rewrite your drivers on the fly?



Or you could write your own!



Fantastic low-level assembly coding book that goes through the beginnings of building your own DOS-like OS

- ▶ But start with the first book *Boot Sector Games* if you're interested ; -)
- ▶ Shows you how to implement Snake!

Might even convince you assembly programming is fun?

- ▶ Or at least make you comfortable with the coding should you take COMSM0049 ~~Systems Software Security Hacking~~
- ▶ Seriously... take me and Sana's unit... we break stuff!

And many more!

OSs are fun

Try something weirder!

Other OSs I haven't mentioned but you should try!

Redox UNIX but rewritten by people who really like Rust

Mezzano Lets rebuild the Lisp machines!

Emacs Comes with a bad text editor

FreeMint ATARI ST OS brought back

SeL4 An OS that has been proven correct and bug free

